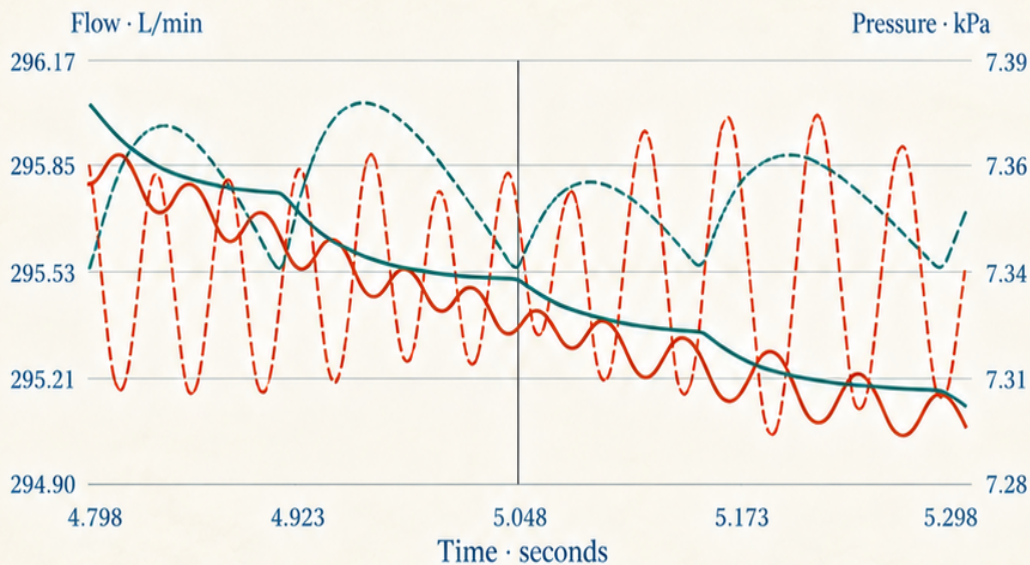


NAVIER-STOKES

A NON-EXPLOSIVE ALTERNATIVE

September 11, 2026



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Navier–Stokes: A Nonexplosive Alternative

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Edition note: the cover retains its original 11 September 2026 artwork date. This edition revises the 20 September rewrite; the accepted cover was preserved on 5 October 2026.

Full rewrite · Locked edition 1.1 · 5 October 2026

“A fundamental problem in analysis is to decide whether such smooth, physically reasonable solutions exist for the Navier-Stokes equations.” — Charles L. Fefferman [1]

Abstract

This paper develops a signed vorticity ledger for the three-dimensional incompressible Navier–Stokes equations and connects it to an executable MATHIC score. Positive stretching, compressive stretching, viscosity and the signed residual remain visible. For externally forced flow, the additional channel is the pairing of vorticity with the curl of the force. The resulting One-Tension identity reduces exactly to the unforced formula; spatially addressed balances retain the transfer terms introduced by localization.

The mathematical account is accompanied by a historical finite calibration, a 35-step diagnostic corridor, exact symbolic field fixtures and a checked viscosity-rescaling subset. The new score executes declared operations from supplied fields and committed predecessor states. These records establish specific algebraic and computational results. The remaining analytic task is a resolution-uniform bound strong enough to enter classical continuation theory.

Claim boundary: the local accounting identity is established; global three-dimensional regularity remains open.

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Closing records: Reference activation index; Claim map; Manuscript status.

1. The problem, and which branch this paper pursues

For an incompressible fluid, let velocity be $u(x, t)$, pressure $p(x, t)$, viscosity $\nu > 0$, and external force $f(x, t)$. The equations are

$$\partial_t u + (u \cdot \nabla) u = -\nabla p + \nu \Delta u + f, \nabla \cdot u = 0.$$

The official problem distinguishes global smoothness in the unforced setting from breakdown under the permitted forcing hypotheses [1]. Those quantifiers matter. A forced singular flow would not show that every unforced flow becomes singular, and an identity valid along a smooth flow would not show that smoothness lasts forever.

The question pursued here is whether the stretching account can be made sufficiently explicit to support a uniform estimate for the unforced smooth-flow branch. We also formulate the account with forcing, so that the same definitions can inspect supplied forced fields without silently discarding their input. Mathematics is occasionally generous enough to let both sides of an argument be interesting.

The original NS ANEA paper supplied the unforced ledger, a local One-Tension theorem and a finite diagnostic. The present formulation brings those elements together with the forced-vorticity extension, explicit localization transfer and the R3 executable score. MATHIC denotes the formula catalogue, typed operations, state transitions and associated evidence. TRYX supplies the Value–Perception–Action cycle; ENAID supplies the addressed organizational framework. Their implementation is described where it becomes operational, rather than being asked to carry an argument by name alone.

The source literature is the fifteen-entry bibliography retained from the locked edition. Its Clay status entry records an access date of 11 September 2026 [2]. The comparison with the September preprint [3] uses the admitted source and its recorded equation locator; it is not an announcement of a later institutional decision. Section 8 states precisely which part of that comparison was computed.

Reference activation. [1] and [2] establish the official problem and historical status context; [3] supplies the forced-construction comparison. None verifies the local theorem proved in §5.

2. The trouble with stretching

“The phenomenon of vortex stretching has frustrated all attempts to date” — Terence Tao [4], discussing vorticity transport.

The difficulty becomes visible after taking curl. Define

$$\omega = \nabla \times u, S = \frac{1}{2}(\nabla u + \nabla u^T).$$

Then the forced vorticity equation is

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega + \nabla \times f.$$

When $f = 0$, the last term vanishes and the original unforced equation is recovered. The symmetric strain S controls the scalar stretching contribution $\omega^T S \omega$. The antisymmetric part of the velocity gradient annihilates the vorticity vector, so stretching is governed by the symmetric strain. Its contribution to the scalar product is zero.

Viscosity damps vorticity gradients. Stretching can amplify vorticity when its direction aligns with an extensional strain direction. The trouble with stretching is not that it stretches; it is that three dimensions give it somewhere to go.

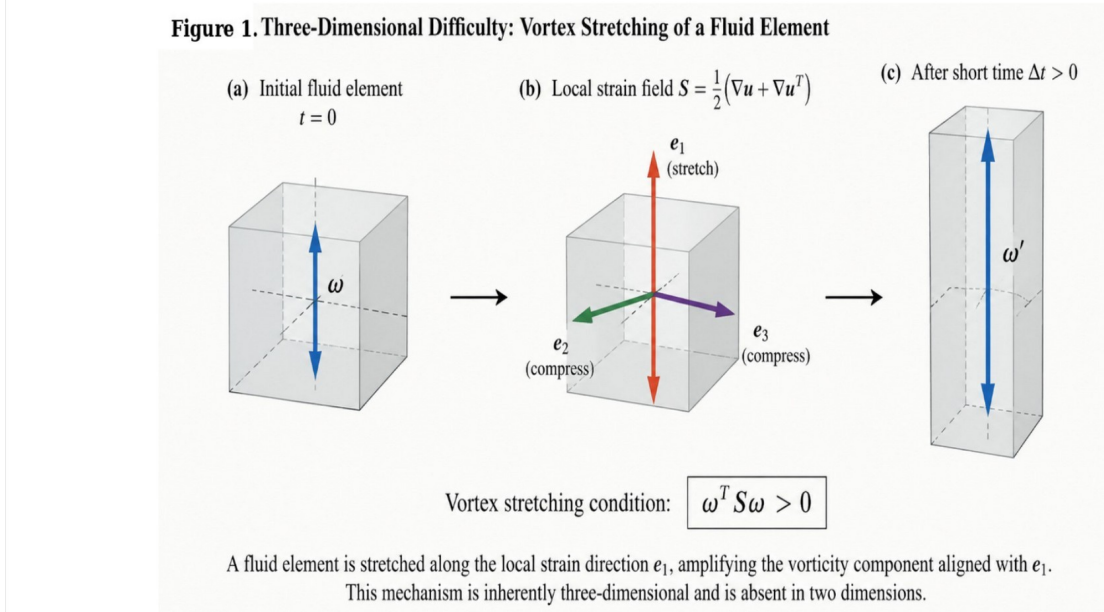


Figure 1. Local vortex-stretching geometry, retained from locked NS ANEA V2. The illustration explains alignment and amplification; it is not a computed fluid solution.

Constantin and Fefferman showed that suitable coherence in the direction of vorticity can deplete the nonlinear mechanism under their hypotheses [5]. This gives geometric organization a precise role in regularity theory. It does not supply those hypotheses for the construction in this paper.

Beale, Kato and Majda identify a vorticity criterion for breakdown of smooth three-dimensional Euler solutions [6]. Euler and Navier–Stokes have different equations and regularity theories; the reference is an analogy about what vorticity can reveal, not a transferred Navier–Stokes theorem. If smoothness fails, vorticity is not a decorative witness.

Viscosity is quite willing to smooth things over. Vortex stretching is less interested in reconciliation.

Reference activation. [4] is exposition, [5] is a geometric regularity result under stated assumptions, and [6] is an Euler breakdown criterion. Their roles remain distinct.

3. A balanced account: the classical enstrophy ledger

Take either a periodic domain Ω , or $\mathbb{R}^3$ with sufficient decay to justify the integrations below. Assume a smooth incompressible solution, finite displayed integrals and differentiation under the integral sign. A bounded-domain treatment requires boundary conditions that actually remove the

relevant fluxes; otherwise those terms must be retained. All unqualified integrals in the ledger are over Ω .

Multiplying the vorticity equation by ω , integrating and using incompressibility gives

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla \omega\|_2^2 = \int \omega^T S \omega dx + \int \omega \cdot (\nabla \times f) dx.$$

Write $E(t) = 1/2 \|\omega(t)\|_2^2$. The left side contains its derivative and viscous dissipation. On the right, stretching and forcing can each add or remove enstrophy. Keeping their signs is essential. Set

$$q = \omega^T S \omega = q_+ - q_-, q_+ = \max(q, 0), q_- = \max(-q, 0).$$

and define

$$A^+ = \int q_+ dx, A_{stretch}^- = \int q_- dx, D = \int |\nabla \omega|^2 dx,$$

The force channel is

$$F_\omega = \int \omega \cdot (\nabla \times f) dx.$$

Split this integrated pairing into its two signs:

$$F_\omega^+ = \max(F_\omega, 0), F_\omega^- = \max(-F_\omega, 0), F_\omega = F_\omega^+ - F_\omega^-.$$

The stretching split occurs pointwise before integration; the force split occurs after integration. These are different operations and cannot be interchanged merely because their notation looks similar. A large force magnitude alone says nothing about the sign of its vorticity pairing. A smooth gradient force has zero curl and can therefore have zero F_ω without being zero as a field.

The forced signed Action is

$$\Delta A_f = A^+ - A_{stretch}^- - \nu D + F_\omega^+ - F_\omega^- = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2.$$

For $f=0$, this becomes the original residual

$$\Delta A = A^+ - A_{stretch}^- - \nu D = \frac{dE}{dt}.$$

The name “Action” identifies a place in the accounting protocol. It does not add a physical term to the Navier–Stokes equations. Positive stretching, compression, viscosity and the surviving remainder are kept separately visible so that later estimates have an explicit target.

Leray’s weak-solution construction supplies the classical finite-energy baseline [7]. That baseline does not, by itself, settle higher regularity. Constantin’s observation that the mathematical theory “is incomplete and requires cut-offs” [8] is particularly pertinent to any calculation indexed by resolution. Exactness after projection must not be confused with control of the limit.

This sounds modest because it is modest. The moment a bookkeeping definition is asked to do the work of an a priori estimate, the paper has borrowed money from a theorem it does not yet own.

Reference activation. [7] supplies the weak-energy background; [8] motivates keeping approximation defects visible.

4. Addressing the interaction: VPA as an audit protocol

The address carried by a comparison is

$$\alpha = (N, x, \text{phase}, \text{orientation}, \text{scale}).$$

The local audit cycle is

$$V \rightarrow P \rightarrow A \rightarrow V'.$$

Value holds the state under examination. Perception performs the declared comparison. Action commits the result, including its residual, and hands that result to the next Value. The address identifies where and at what scale the comparison is made. These are roles in an audit protocol, not new physical dimensions.

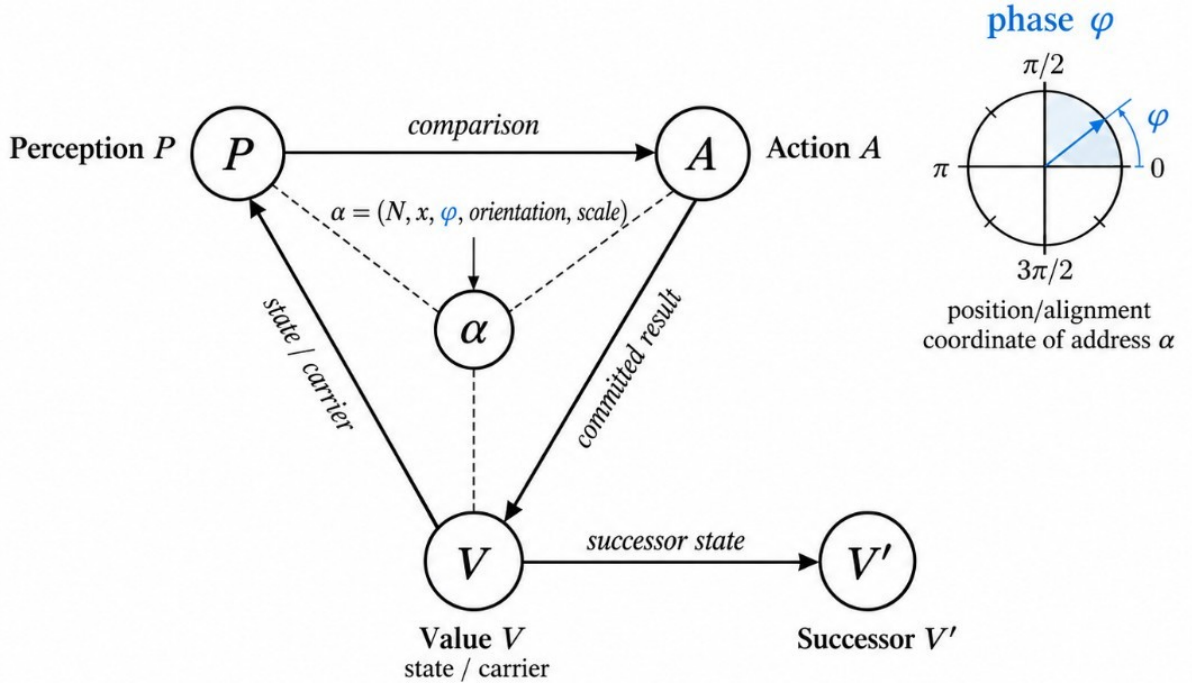


Figure 2. Addressed audit cell. Retained from locked NS ANEA V2. The phase coordinate belongs to the address; the triangular orientation distinguishes the audit roles and does not assert a new physical identification.

The residual belongs to the committed Action. A calculation may not use a partially evaluated result as a successor state before the declared frame commits. Nor may it balance one scale against a convenient term from another address without declaring the relation. Kato's function-space framework [9] illustrates why those declarations must eventually become precise analytic hypotheses. The scale parameter is not impressed by a million successful examples.

4.1 Local balances need their transfer

For a smooth nonnegative physical-space weight $\chi_\alpha(x, t)$, define

$$E_\alpha(t) = \frac{1}{2} \int \chi_\alpha |\omega|^2 dx.$$

The exact localization contribution is

$$L_\alpha = \frac{1}{2} \int (\partial_t \chi_\alpha + u \cdot \nabla \chi_\alpha + v \Delta \chi_\alpha) |\omega|^2 dx.$$

Here $A_\alpha^+ = \int \chi_\alpha \max(q, 0) dx$, $A_{stretch, \alpha}^- = \int \chi_\alpha \max(-q, 0) dx$, $D_\alpha = \int \chi_\alpha |\nabla \omega|^2 dx$, and $F_{\omega, \alpha} = \int \chi_\alpha \omega \cdot \text{curl } f dx$. All integrals use the declared domain.

With weighted stretching, dissipation and force channels, the balance reads

$$\frac{dE_\alpha}{dt} = A_\alpha^+ - A_{stretch, \alpha}^- - v D_\alpha + F_{\omega, \alpha} + L_\alpha.$$

This formula follows from the product rule, transport integration by parts and the weighted viscous integration by parts under the same boundary assumptions. The term L_α keeps the derivatives of the weight. Erasing it changes the equation.

For a Fourier, Galerkin, Littlewood–Paley or filtered address, the transfer or commutator must instead be derived for the chosen operator. The declared account has the form

$$\Delta A_f(\alpha) = A^+(\alpha) - A_{stretch}^-(\alpha) - v D(\alpha) + F_\omega(\alpha) + C_\alpha.$$

Here $C_\alpha = L_\alpha$ for the physical-space weights above. A spectral C_α is not supplied by copying this expression into a new notation. No general spectral commutator estimate is established here.

The executed partition uses the periodic cube $[0, 2\pi]^3$, unit viscosity and

$$\chi_0(y) = 1/2(1 + \cos 2y) = \cos^2 y, \chi_1(y) = 1/2(1 - \cos 2y) = \sin^2 y.$$

For the growing F4 shear flow at time zero, the local transfers are $L_0 = -2\pi^3$ and $L_1 = 2\pi^3$. Their sum vanishes, but each local account needs its own contribution. The weighted balances reconstruct the global quantities only after those transfers are included. An account may balance across the room while something is still being carried between the tables.

Reference activation. [9] supplies the strong-solution and function-space context, not a proof of VPA. The cutoff caution in [8] also applies to the distinction between physical localization and an unproved projected estimate.

5. The One-Tension local theorem

For the unforced residual, write $\Delta A^\pm = \max(\pm \Delta A, 0)$. Then $\Delta A = \Delta A^+ - \Delta A^-$.

Theorem 1 — One-Tension Action Accounting. Let the three unforced channels at the declared comparison be finite and let $\Delta A = A^+ - A_{stretch}^- - \nu D$. Then

$$A^+ + \Delta A^- = A_{stretch}^- + \nu D + \Delta A^+.$$

Proof. Equate the definition of ΔA with its positive-negative decomposition and rearrange. No estimate is used. $\square$

The forced version retains both force signs. Define

$$\Delta A_f^+ = \max(\Delta A_f, 0), \Delta A_f^- = \max(-\Delta A_f, 0).$$

Theorem 2 — Forced One-Tension Accounting. For finite channels defined in §3,

$$A^+ + F_\omega^+ + \Delta A_f^- = A_{stretch}^- + \nu D + F_\omega^- + \Delta A_f^+.$$

Proof. Substitute $\Delta A_f = \Delta A_f^+ - \Delta A_f^-$ into the signed Action definition and move its negative terms to the opposite side. $\square$

If $f=0$, then $\nabla \times f=0$, so

$$F_\omega^+ = F_\omega^- = 0.$$

and

$$\Delta A_f = A^+ - A_{stretch}^- - \nu D = \Delta A,$$

Theorem 2 therefore reduces exactly to Theorem 1. At an address, a nonzero transfer remains part of the signed residual; applying the same algebra requires carrying its sign as well. A vanishing force does not grant permission to drop a nonzero localization term.

The local theorem says that the account can balance without deleting a signed contribution. It does not say that the positive residual vanishes, that stretching is bounded, or that concentration is controlled. Equality is easy to obtain once the remainder is retained. Exhaustion is harder. A balanced account does not imply that nobody is still carrying something.

5.1 What the retained Lean evidence proves

The MATHIC v7.3 algebraic successor imports the locked v7.2 modules and adds sign-separated force channels, forced One-Tension, exact zero-force reduction, transfer retention, neutral-transfer reduction and rejection of a nonzero force channel as an unforced no-op. The retained Section 4 records contain the scalar build and axiom audit for the repaired successor.

These are scalar algebraic results. They do not formalize the PDE integrations by parts, the full localization derivation, spectral commutator estimates or the external blowup proof. The present rewrite preserves that verification record and performs no new Lean run. The relevance of stretching geometry comes from the classical literature [5]; the accounting identities are proved above.

6. The finite fixture: useful evidence, carefully fenced

A fixture should tell us what was exercised and what survived. It should also tell us when to stop reading into the picture.

6.1 The retained finite calibration

The locked MATHIC v6 predecessor records amplitude $A=7$, viscosity $\nu=1$, and the channel values

$$A^+ = 820.87564089412, A_{stretch}^- = 134.87564089412,$$

$$\nu D = 588, \Delta A^+ = 98, \Delta A^- = 0.$$

The corresponding arithmetic is

$$820.87564089412 = 134.87564089412 + 588 + 98.$$

The equality residual is zero while the positive remainder is 98. The calibration therefore illustrates residual retention, not initial exhaustion.

The predecessor also records the reflected update

$$E_{j+1} = \max(E_j + \Delta A_j, 0).$$

The stored 35-step corridor starts from a zero carrier, reaches its recorded maximum at ordinal 17 and returns to zero at ordinal 34. The figure is retained from that record; it is not a newly generated simulation.

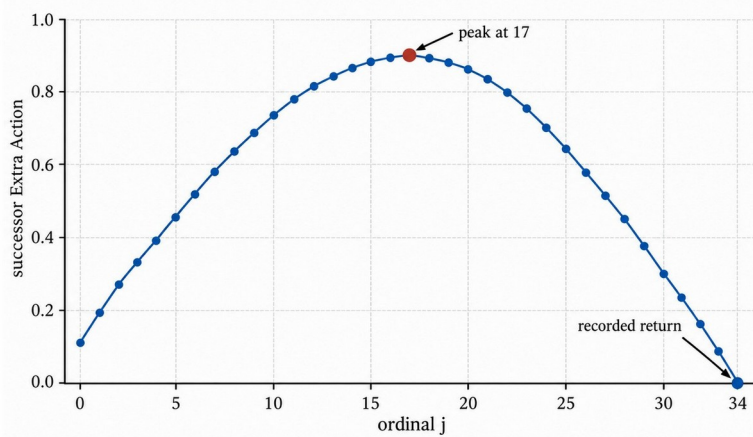


Figure 3. Historical finite corridor from the locked v6 fixture, as reproduced in NS ANEA V2. The plotted successor sequence tests bookkeeping and return behavior. Its vertical scale must not be read as a physical Navier–Stokes enstrophy measurement.

The replay asks legitimate implementation questions: whether signs survive, whether the remainder is carried, and whether clipping occurs at the declared stage. It does not establish a Galerkin limit or show that every continuum flow returns to zero. One good encounter is not a long-term relationship.

6.2 From a saved witness to an executed score

The newer score admits supplied fields and computes or receives

$$\omega = \nabla \times u, S = 1/2 (\nabla u + \nabla u^T), g = \nabla \times f,$$

It then evaluates the signed channels, residual and declared transfer. Phase, orientation, scale and provenance stay attached to the address. The earlier receipt score projected saved witnesses; R3 executes field operations from input expressions and the committed predecessor. A browser chart displays the execution records, while Python performs new admitted inputs.

6.3 Admitted inputs and typed operations

The field carrier records three-component velocity and force, scalar pressure, positive constant viscosity, physical time, the periodic domain and source provenance. The implementation derives vorticity, strain and curl-force from the input expressions. It admits the restricted arithmetic and elementary-function grammar implemented in the evaluator. Its supported fixture interval is $0 \leq t \leq 1$. Smoothness, integrability and the declared carrier remain mathematical obligations; this implementation is not an unrestricted PDE solver.

The score first checks incompressibility, the momentum equation and symbolic periodicity. Subsequent operations derive the vorticity equation, integrate stretching and dissipation, integrate force pairing, split its sign, form signed Action, and compare One-Tension and the enstrophy derivative. Address operations then bind declared weights, compute local channels, retain transfer and reconstruct the global balance. The external input has a separate scaling carrier; unavailable absolute fields are not replaced with zeros.

6.4 Four synchronized component scores

Component	Executed role	Returned evidence
Atomic	Admission, differential fields and channel contributions	Typed fields, channel values and source bindings
Micro	Spatial weights, local channels and retained transfer	Local balance records and transfer guards
Macro	Signed Action, global balances and reconstruction	Exact residuals and reconstructed quantities
Squadron	Cross-scale comparison and mutation audit	Localized mismatch or consistency record

The table is the static four-staff explanation. Construction proceeds from field operations through local and global quantities; Squadron compares the available evidence. It cannot create a missing field or repair another staff's result.

A component measure has one typed note or rest, denoted T0[1|1], passing through Value, Perception and Action. Value retains the committed predecessor. Perception evaluates the declared operation. Action commits the complete frame only after all four component returns are

assembled. Each component reads the same predecessor, so a partially evaluated member cannot become another member's input within that frame.

An explicit Sigma relation declares the four-member co-evaluation frame. Equal ordinal positions alone do not establish it. The host executes a serialized bounded representation; this makes no claim about physically simultaneous processors. Physical PDE time, ordinal progression and display timing remain distinct. REST preserves a non-operation's score position. MUTE preserves a missing or inapplicable return with its reason. HOLD stops progression at an invalid input or mismatch.

Every performance has 18 frames and 72 component returns. Each carries the source-note binding, typed address, operation, VPA record, state hashes and result. The next frame consumes the actual committed successor. No reflection, antipodal cancellation, swing or fold operation is used by this specialization. The terminal carrier is NSFieldState, not a scalar slogan or a TemplexOne claim.

6.5 Exact fixture results

For F0, F1, F2 and F4, the velocity has the form $u = (a(t)\sin y, 0, 0)$, pressure is zero and the force is $f = ((a'(t) + a(t))\sin y, 0, 0)$. Viscosity is one. F0 uses $a = e^{-t}$, F1 and F4 use $a = e^t$, and F2 uses $a = e^{-2t}$. F3 uses the F0 velocity with a nonzero gradient force and a matching pressure. The archived inputs specify those fields exactly.

Input and time	Enstrophy E	Force pairing $F\omega$	Signed Action ΔAf
F0 at 0	$2 \cdot \pi^3$	0	$-4 \cdot \pi^3$
F1 at 0	$2 \cdot \pi^3$	$8 \cdot \pi^3$	$4 \cdot \pi^3$
F2 at 0	$2 \cdot \pi^3$	$-4 \cdot \pi^3$	$-8 \cdot \pi^3$
F3 at 0	$2 \cdot \pi^3$	0	$-4 \cdot \pi^3$
F4 at 0	$2 \cdot \pi^3$	$8 \cdot \pi^3$	$4 \cdot \pi^3$
F4 at 1/2	$2 \cdot e \cdot \pi^3$	$8 \cdot e \cdot \pi^3$	$4 \cdot e \cdot \pi^3$

All listed field results have exactly zero global balance and One-Tension residuals. F4 also reconstructs enstrophy, its derivative, dissipation, force pairing, stretching and signed Action from the two weighted parts. The extra F4 evaluation at time 1/2 produces changed values, demonstrating that the score is not confined to displaying the original saved fixture values.

These shear fixtures have zero stretching density. Their success verifies the declared operations and input cases; it does not establish arbitrary nonzero-stretching behavior, all possible input expressions or universal continuum control. The old formula definitions remain broader conditional mathematics, while the computation evidence remains attached to its tested carrier.

6.6 Validation and mechanism controls

The earlier receipt score projected saved witnesses. R3 computes field operations from the source input and committed state; its execution path does not read saved result JSON files. Validation

freshly executes the supplied input and compares the resulting operations, states, outputs, provenance and checksums with the submitted performance.

The repaired standalone receipt validator checks nested digests separately from semantic agreement. Missing or changed checksums fail; re-signing a false semantic statement does not make that statement valid. The archived test run accepts six baseline receipt performances and passes eight integrity controls. Field tests pass 29 score/transcript negative controls and six execution checks. Four controls for unused swing, reflection or cancellation operations have explicit not-applicable dispositions. These counts describe different test categories and are not independent theorem counts.

Mechanism checks include force deletion, reversal of the negative-force sign, erased local transfer, invalid PDE input, lost provenance, premature commit and an incorrect repeat source. Faults are located to a score component, measure or receipt field. The fresh comparison uses the separately organized existing evaluator, which shares SymPy and primitive calculus routines; it is regression evidence rather than a second proof system.

Trust is welcome. Receipts are forever.

The fixture evidence belongs to the exact declared inputs, evaluator and transcript versions. No external reference is used to validate the old synthetic corridor or to turn these new symbolic examples into a general PDE theorem.

7. Smoothness is a long-term relationship

“The discovery of any new bound, stronger than that provided by the energy ... would have the significance of a major event.” — Sergiu Klainerman, quoted by Terence Tao [10]

The local theorem tells us where the missing work lives. To enter global regularity theory, the account needs a bound that survives refinement. The same finite identity can hold at every resolution while its useful constants grow beyond control.

A schematic target for the unforced flow is

$$|\int (\omega \cdot \nabla) u \cdot \omega \, dx| \leq (1 - \varepsilon) \nu \|\nabla \omega\|_2^2 + C \|\omega\|_2^2,$$

with $\varepsilon > 0$ and C controlled for the relevant domain and data, uniformly in the approximation. If this bound held for the smooth solution class under consideration, the enstrophy equation would imply

$$\frac{d}{dt} \|\omega\|_2^2 \leq 2C \|\omega\|_2^2.$$

Grönwall would then control enstrophy on finite time intervals. The first inequality is a research target, not a result proved here. Its derivation and the required approximation and continuation steps remain obligations; the estimate cannot be borrowed from the conclusion.

Serrin gives explicit spacetime-integrability conditions for regularity [11]. Escauriaza, Seregin and Šverák establish the critical $L_t^\infty L_x^3$ endpoint for the unforced Cauchy problem [12]. Foias and

Temam describe Gevrey regularity, providing a language for analyticity and spectral decay [13]. These results give the program a destination. Their hypotheses are not supplied by a successful fixture.

The open work has four linked parts:

Control the positive-stretching tail uniformly as resolution increases.

Derive and bound the projection or filter commutator rather than hiding it in an address.

Obtain coercive control with constants that survive the continuum passage.

Deduce a recognized continuation criterion for the specified solution class.

At some point the argument has to pay its debt in a norm that survives refinement.

Caffarelli, Kohn and Nirenberg prove partial regularity for suitable weak solutions [14]. That conclusion restricts the possible singular set; it does not eliminate it. Caffarelli’s interview description, “A singularity appears and disappears, so if they exist they have a minimal effect” [15], gives an accessible physical image of the distinction. Small is not the same as absent.

Research target — Uniform Action Control. Derive an estimate for positive stretching and approximation defects, uniform across the declared resolutions, that implies a classical regularity criterion for the unforced three-dimensional equations. **Status: open.**

Reference activation. [10] supplies the discussion of scale difficulty and the attributed observation; [11]–[13] supply distinct regularity criteria and smoothness frameworks; [14] gives partial regularity; [15] supplies commentary rather than a theorem.

8. Relation to the September 2026 forced-blowup construction

The admitted September 2026 preprint [3] reports a forced construction with smooth compactly supported forcing, zero initial velocity, bounded kinetic energy and unbounded velocity near a finite terminal time. Its theorem is a source claim used for comparison. The present paper does not independently verify its full construction or assert a later change to Clay’s official status.

The mathematical objects overlap: nonlinear amplification, viscosity, scale, phase and residual corrections. The quantified questions differ. The smooth-flow program asks what can be controlled when the force vanishes; the external construction describes what can happen with a deliberately supplied force. The forced problem is allowed to have a very bad day without proving that the unforced problem must join it.

Given supplied fields, the signed ledger can record the following channels.

External object	Ledger channel
Velocity and pressure	input carriers u, p
Smooth external force	$f, \text{curl}(f), F_{\omega+}, F_{\omega-}$
Vorticity amplification	$q, A+, A\text{-stretch}$

External object	Ledger channel
Viscous action	νD
Shrinking scales	address scale and localization witness
Oscillatory phases	phase/orientation address
Residual corrections	force provenance and transfer/commutator record
Bounded kinetic energy	separate kinetic-energy observable
Unbounded velocity	separately supplied singularity observable

Omission of this paper's terminology from another publication says nothing about what its authors can calculate. Conversely, this ledger's ability to organize supplied quantities does not recreate a 166-page construction.

8.1 The executed rescaling subset

The admitted F5 input uses the viscosity rescaling in [3, equation (10.22), printed page 124]. Its transformation is

$$u_v(x, t) = \sqrt{v} u_1(x / \sqrt{v}, t), p_v(x, t) = v p_1(x / \sqrt{v}, t), f_v(x, t) = \sqrt{v} f_1(x / \sqrt{v}, t).$$

The parameter $v > 0$ is fixed viscosity; time is unchanged. R3 symbolically checks fourteen relations, including chain rules, the transformed momentum equation, the volume Jacobian and positive-scaling sign splits. With the transformed integration domain, the recorded enstrophy and ledger channels scale as $v^{3/2}$, dissipation D as $v^{1/2}$, and kinetic energy as $v^{5/2}$.

Quantity group	Evidence state
Supplied scaling relations and source locator	Source-pinned input
Chain rules, scaling factors and ledger relations	Fourteen exact symbolic checks passed
Absolute external u , p and f	Not computed
Absolute channel values and address transfers	Not computed
Full external construction or independent blowup proof	Not performed or claimed

The retained source-admission record checked the theorem statement and rescaling locator for the earlier execution work. This is verification of the cited scope and formula, not independent validation of the full preprint proof.

The force channel can therefore be mapped without claiming that the missing absolute fields have been computed. Scaling exponents tell us how an admitted field would transform; they do not supply the field itself.

9. Discussion: what the framework contributes and what it still owes

The contribution is an explicit account of vorticity growth whose terms remain identifiable through local algebra, spatial localization and a bounded executable score. The force term has a sign, transfer has a destination, and every computation carries its input and source binding. The historical corridor remains available as a semantic replay; R3 adds actual field operations for the declared symbolic carrier.

That organization matters because global regularity asks for more than a sum that balances. Tao’s discussion of supercriticality [10] explains why energy information weakens at finer scales. Constantin’s cutoff warning [8] reaches the same difficulty from the approximation side. Repeating a local identity at more addresses does not by itself control the defect in passing to the limit.

Figure 4 compares two controllers in a reduced synthetic pump model. Its traces are regenerated from the preserved executable controller, rather than inferred from an image. These calculated outputs are not hardware measurements and do not enter the regularity argument.

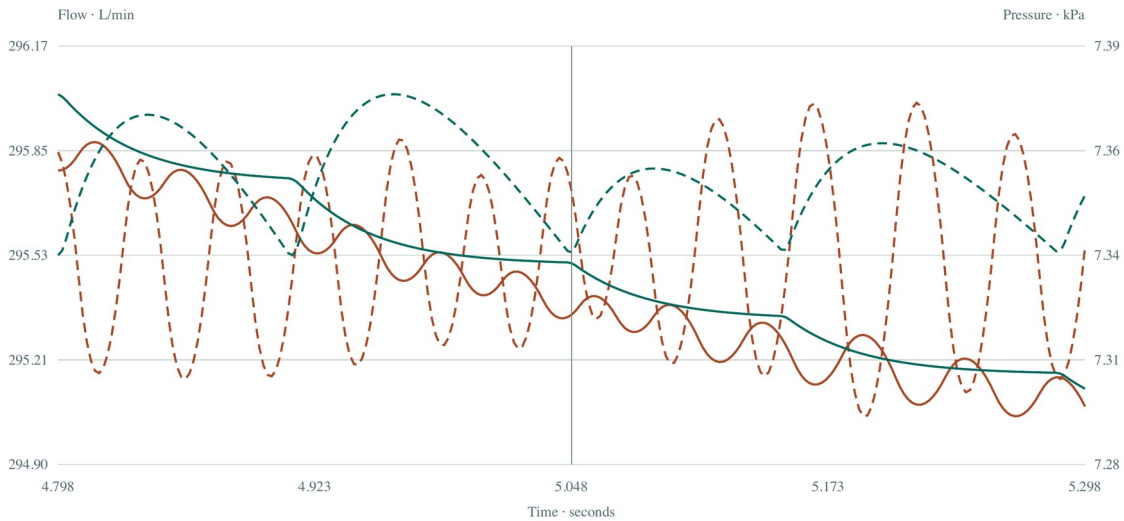


Figure 4. FE6B flow-controller model comparison. Rust: independent PI comparator; teal: signed NS ANEA controller. Solid: flow (L/min, left axis); dashed: pressure (kPa, right axis). Requested flow 400 L/min; peak restriction parameter 0.65; timestep 0.002 s; window 4.798–5.298 s. The PI implementation references a MathWorks method but does not run a MathWorks product. Source and numerical series: companion evidence/NS_FLOW_CONTROLLER.html and figure4-data.json.

The next analytic steps are concrete: place the signed channels in a Galerkin or Littlewood–Paley setting; compute the associated commutator; seek depletion or coercivity uniform in resolution; and test the resulting estimate against a classical continuation criterion. The mountain is visible. The rocks still matter.

10. Methodological note on human-AI work

This work was developed through iterative human–AI collaboration. AI assisted with literature retrieval, finite calculations, comparison of formulations, counterquestions, source reconciliation and manuscript preparation. Mathematical responsibility remains human. The record distinguishes definitions from theorems, finite checks from continuum conclusions, and imported statements from results derived here.

A generated sentence does not become evidence by arriving in mathematical notation. Computation is useful when its input, operation and returned result can be inspected; prose is useful when it lets another reader follow the same distinction. The formulas, execution records and reproduction instructions are supplied for that purpose.

AI is a kite. Imagination creates lift; the human hand supplies the tension that gives flight direction.

The retained Lean record, the R3 symbolic tests and the historical corridor have different proof weight. No new kernel run or fresh full external-proof replay is claimed by this rewrite.

11. References

The numbering and bibliographic metadata below retain the locked NS ANEA V2 reference set. Historical access dates remain historical. Source descriptions identify the role and status recorded in that edition; the evidence register separately identifies what was checked for the present integration.

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- [2] Clay Mathematics Institute, “The Millennium Prize Problems” and “Navier-Stokes Equation,” current status pages, accessed 11 September 2026.
- [3] OpenAI, “Finite Time Blowup for Navier-Stokes,” preprint, 8 September 2026, 166 pp.; accompanied by a Lean formalization. Theorem 1.1 constructs smooth compactly supported forcing and finite-time unbounded velocity from zero initial velocity with bounded kinetic energy. Retained comparison locator: equation (10.22), printed page 124.
<https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>
- [4] T. Tao, “254A, Notes 3: Local well-posedness for the Euler equations,” *What’s New*, 9 October 2018. Expository source for the quoted observation about vortex stretching.
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- [7] J. Leray, “Sur le mouvement d’un liquide visqueux emplissant l’espace,” *Acta Mathematica* 63 (1934), 193-248. DOI: 10.1007/BF02547354.
- [8] P. Constantin, “Near Identity Transformations for the Navier-Stokes Equations,” 2002 manuscript. Used here for the quoted observation that the mathematical theory remains incomplete and requires cut-offs.
- [9] T. Kato, “Strong L^p -Solutions of the Navier-Stokes Equation in R^m , with Applications to Weak Solutions,” *Mathematische Zeitschrift* 187 (1984), 471-480. DOI: 10.1007/BF01174182.
- [10] T. Tao, “Why global regularity for Navier-Stokes is hard,” *What’s New*, 18 March 2007. Includes the quoted Klainerman observation and an exposition of supercriticality.
- [11] J. Serrin, “On the Interior Regularity of Weak Solutions of the Navier-Stokes Equations,” *Archive for Rational Mechanics and Analysis* 9 (1962), 187-195. DOI: 10.1007/BF00253344.
- [12] L. Escauriaza, G. Seregin, and V. Šverák, “ $L^{3,\infty}$ -Solutions of the Navier-Stokes Equations and Backward Uniqueness,” *Russian Mathematical Surveys* 58 (2003), no. 2, 211-250. DOI: 10.1070/RM2003V058N02ABEH000609.

- [13] C. Foias and R. Temam, “Gevrey Class Regularity for the Solutions of the Navier-Stokes Equations,” *Journal of Functional Analysis* 87 (1989), no. 2, 359-369. DOI: 10.1016/0022-1236(89)90015-3.
- [14] L. Caffarelli, R. Kohn, and L. Nirenberg, “Partial Regularity of Suitable Weak Solutions of the Navier-Stokes Equations,” *Communications on Pure and Applied Mathematics* 35 (1982), no. 6, 771-831. DOI: 10.1002/cpa.3160350604.
- [15] Oden Institute for Computational Engineering and Sciences, “Caffarelli explains role in understanding Navier-Stokes Equations,” interview/news feature quoting Luis Caffarelli, 2015.

Appendix A. Formula progression and evidence crosswalk

The progression is classical PDE → addressed ledger → local identity → finite execution → open analytic promotion. It replaces the locked paper’s standalone unforced formula insert with an integrated formula summary and the retained F01–F18 execution map. The order below follows causal execution: F04 compares energy only after the signed channels have been assembled.

Formula / Frame	Definition or operation	Section
F01 / 0	$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{f}; \nabla \cdot \mathbf{u} = 0$	§1, 6.3
F02 / 1	$\boldsymbol{\omega} = \text{curl } \mathbf{u}; \mathbf{S} = (\nabla \mathbf{u} + \nabla \mathbf{u}^T)/2; \mathbf{q} = \boldsymbol{\omega}^T \mathbf{S} \boldsymbol{\omega}$	§2, 6.2
F03 / 2	$\partial_t \boldsymbol{\omega} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \Delta \boldsymbol{\omega} + \text{curl } \mathbf{f}$	§2
F05 / 3	$A^+ = \int \max(\mathbf{q}, 0); A^- \text{-stretch} = \int \max(-\mathbf{q}, 0); D = \int \nabla \boldsymbol{\omega} ^2$	§3
F06 / 4	$F\boldsymbol{\omega} = \int \boldsymbol{\omega} \cdot \text{curl } \mathbf{f}$	§3
F07 / 5	$F\boldsymbol{\omega}^+ = \max(F\boldsymbol{\omega}, 0); F\boldsymbol{\omega}^- = \max(-F\boldsymbol{\omega}, 0); F\boldsymbol{\omega} = F\boldsymbol{\omega}^+ - F\boldsymbol{\omega}^-$	§3
F08 / 6	$\Delta A\mathbf{f} = A^+ - A^- \text{-stretch} - \nu D + F\boldsymbol{\omega}^+ - F\boldsymbol{\omega}^-$	§3
F09 / 7	$\Delta A\mathbf{f}^+ = \max(\Delta A\mathbf{f}, 0); \Delta A\mathbf{f}^- = \max(-\Delta A\mathbf{f}, 0)$	§5
F10 / 8	$A^+ + F\boldsymbol{\omega}^+ + \Delta A\mathbf{f}^- = A^- \text{-stretch} + \nu D + F\boldsymbol{\omega}^- + \Delta A\mathbf{f}^+$	§5
F04 / 9	$E = \frac{1}{2} \int \boldsymbol{\omega} ^2; E' = A^+ - A^- \text{-stretch} - \nu D + F\boldsymbol{\omega}$	§3, 6.5
F11 / 10	$\mathbf{f} = 0 \Rightarrow \text{curl } \mathbf{f} = 0 \Rightarrow F\boldsymbol{\omega}^+ = F\boldsymbol{\omega}^- = 0$	§5
F12 / 11	$F\boldsymbol{\omega}^+ = F\boldsymbol{\omega}^- = 0 \Rightarrow \Delta A\mathbf{f} = A^+ - A^- \text{-stretch} - \nu D$	§5
F13 / 12	$\alpha = (\mathbf{N}, \mathbf{x}, \text{phase}, \text{orientation}, \text{scale})$	§4
F14 / 13	$E\alpha = \frac{1}{2} \int \chi \alpha \boldsymbol{\omega} ^2; L\alpha = \frac{1}{2} \int (\partial_t \chi \alpha + \mathbf{u} \cdot \nabla \chi \alpha + \nu \Delta \chi \alpha) \boldsymbol{\omega} ^2$	§4.1
F15 / 14	$\Delta A\mathbf{f}(\alpha) = A^+ \alpha - A^- \text{-stretch}, \alpha - \nu D \alpha + F\boldsymbol{\omega}, \alpha + C\alpha$	§4.1
F16 / 15	$C\alpha = 0 \Rightarrow \text{transfer-free addressed ledger}$	§4.1, 6.5
F17 / 16	Other channels fixed: $F\boldsymbol{\omega} \neq 0 \Rightarrow \Delta A\mathbf{f} \neq \Delta A\mathbf{f} - F\boldsymbol{\omega}$	§6.6
F18 / 17	$u\mathbf{v} = \sqrt{\nu} \mathbf{u}_1(\mathbf{x}/\sqrt{\nu}, t); p\mathbf{v} = \nu p_1(\mathbf{x}/\sqrt{\nu}, t); f\mathbf{v} = \sqrt{\nu} \mathbf{f}_1(\mathbf{x}/\sqrt{\nu}, t)$	§8.1

The historical companion JSON is listed in the archived manifest. It records per-fixture frame checksums, operation types, source definitions, hypotheses and verification scope. The four-component explanation is in §6.4. The archive retrieval instructions below identify the historical package; its execution claims have not been rerun for this edition.

Appendix B. Reproduction and assumptions

Historical R3 reproduction requires the archived rewrite packet in TRYX-Relay/TRYX-MATHIC at commit 31d46ba2fdc404ace23cd7b59b8d46785b484b61. Retrieve archive/navier-stokes/version-2/paper-rewrite/260920/NS_ANEA_FULL_REWRITE_PACKET_260920.zip and its sibling MANIFEST.json. Repository access is required. The companion evidence/ARCHIVED_R3_MANIFEST.json records the original per-file SHA-256 values, including mathic/field_score.py and mathic/inputs/F4.json. Verify the extracted files before execution. Use the archived requirements.txt; the historical environment specifies Python 3.12 and SymPy 1.14.0. From the extracted mathic directory:

```
python field_score.py inputs/F4.json /tmp/F4.json
python validate_performance.py field field-score.json \
  /tmp/F4.json --input inputs/F4.json
python validate_performance.py receipt receipt-score.json \
  receipt-results/F4.json
```

The field validator executes the admitted input afresh. The receipt validator verifies the bounded projection score and its nested digests. Their scopes are distinct. The regression suite is test_field_score.py; run it in a working copy because it regenerates new score and performance records. Historical fixture and Lean results are retained rather than automatically replayed for a typesetting change.

Source admission uses pinned bodies, versions and hashes. Canonical semantic JSON uses sorted keys and compact UTF-8 encoding; exact expressions are strings. Each checksum omits only its own field while preserving nested digests. The complete transcript binds component receipts, braid, input and terminal state. SHA-256 detects mismatched bytes under the declared source trust; it is not a digital signature or proof of a PDE theorem.

The archived manifest lists the declared inputs, seven R3 performances, field-conformance-tests.json, field-chart-checks.json, scalar build/axiom logs and source records. These are historical reported results; neither the R3 suite nor Lean was rerun for this edition. The companion bundle provides the retrieved manifest and the separately verified controller source and figure data. It does not redistribute the full historical R3 packet. A nonzero-stretching fixture and the resolution-uniform estimate remain open validation and analytic tasks.

For the global ledger, assume smoothness, incompressibility, sufficient integrability and the declared periodic or vanishing-flux setting. The force channel requires the displayed curl and pairing. For the spatial ledger, assume a smooth weight and the integrations by parts used in §4.1. Nonnegative weights are used when interpreting weighted channel parts as nonnegative, and a partition of unity is required for the implemented reconstruction. For the F5 scaling, viscosity is positive and the integration domain is transformed consistently. Regularity promotion additionally requires an independently proved uniform estimate and a justified continuum passage.

The retained H01–H17 source hypotheses and formula provenance remain in the accompanying MATHIC source. The interface accepts only its declared grammar and carrier. Neither syntax admission nor a digest supplies missing analytic assumptions.

Reference activation index

[1]–[3] — §§1 and 8. Official problem boundary and forced-construction comparison.

[4]–[6] — §2. Stretching, geometric depletion and Euler breakdown analogy.

[7]–[8] — §§3 and 4.1. Weak-energy baseline and approximation caution.

[9] — §4. Function-space precision and strong-solution context.

[10]–[15] — §7. Scale difficulty, regularity criteria, partial regularity and commentary.

Claim map

Statement	Status and location
Unforced and forced One-Tension	Exact scalar identities, proved in §5; retained scalar Lean evidence
Global forced enstrophy account	Derived under the stated smoothness and boundary assumptions, §3
Physical-space addressed balance	Transfer retained under stated assumptions, §4.1
Historical calibration and corridor	Retained finite record, §6.1
R3 exact field execution	Declared periodic fixtures and fresh F4 time input, §6
External viscosity rescaling	Fourteen symbolic relations checked in the admitted subset, §8.1
Full external singular construction	Not replayed or independently proved here
Uniform Action Control	Open, §7
Global three-dimensional unforced regularity	Open; no theorem promotion from the finite evidence

Manuscript status

Locked edition 1.1, 5 October 2026, succeeding the 20 September 2026 rewrite. This edition corrects equation rendering, vorticity wording, figure numbering, figure provenance and reproduction instructions. It preserves the accepted splash artwork and previous locked publications. The signed accounting identities are established under the stated assumptions; historical execution claims retain their original scope. Nonzero-stretching execution coverage and the uniform analytic estimate remain open.